



COLLECTING SOLID WASTE FROM POLLUTED WATERS USING 'WASTE SHARK MEGA BOX' ROBOT

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Collecting Solid Waste from Polluted Waters using ‘Mega box’ Robot

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INTRODUCTION

Plastic products have played significant roles in protecting people during the COVID-19 pandemic. The widespread use of personal protective gear (PPE), and the popularity of online shopping have created a massive disruption in the supply chain and waste disposal system. Robotics, on the other hand, helps to solve complex tasks using engineering and science to solve one of the biggest problems, pollution. So, the thought of making a waste collecting robot that would pick up trash like the “Waste Shark” with “LiDAR” to increase both safety (collision avoidance) and enhanced data captured from the environment that works in all bodies of waters came. But the budget is also important. Then, the thought of coming up with a waste collecting robot that would collect trash manually and automatically bring it back, but the most difficult part is the system that would return to the starting position also came. So, the final product is a ‘Mega-box’ waste collecting robot that is RC controlled and would have a strong and cost-effective motor using the cheapest materials that would overcome the current of some waters and collect wasted PPE’s and other trash.

This waste collecting robot can be used in most calm bodies of water. So, if the robot is used when the waves or tides aren’t too big, the use of this robot will be much easier. One of the limitations of this waste collecting robot is its battery lifespan. Since this robot machinery will use a controller for it to move around water, it will require the use of a battery. This means that the robot can be only used for a certain period and after using it for some time, the battery should be replaced because it may have been in contact with water for quite a bit. Another limitation that the robot has is its weight capacity. Since the current item used for the body of the robot is a mega box which is made from plastic, the amount of trash that it can collect will be limited but it will still depend on what trash it will pick up.

This Investigatory Project's goal is to reduce waste in bodies of water. Another objective is to build a low-cost robot that can be built as a DIY project in the comfort of anyone's home or workshop and yet perform well. During evaluation, this research has given us **the** following problems:

- What conditions can the **WSM (waste shark megabox)** perform to its maximum potential in terms of (a) quality of water sources based on data collected from sensors or water monitoring system (b) waste collection



- How much trash/pollution can the megabox carry in one session while in water?
- How long would the megabox function under heavy payloads - requiring maximum speed due to too many trash/pollutions inside the net?

Water is one of the important things people need to survive. This study aims to help solve the problem of water pollution, by starting on locations such as on water since it is harder to clean up in those locations than on land. **This study** also enhances the creativity of Computer Science and Engineering.

MATERIALS AND METHODS

Materials

The materials that the researchers need for this study are the mega box (66.5 cm x 45 cm) (1), fish net (1), metal propeller (1), and servo (1). The study also needed Shafts stainless steel (1), brushless motor (1), ESC (1), and 4s/5s battery (1). Last is the receiver (1), the research needs a receiver for the robot to work.

Methods

Finding ways to clear bodies of water.

Rivers for example are a vast expanse of water, polluted by human wastes and trash. The researchers need to find ways to clean it up, like we would care for our own lawn or backyard.

Researching about the waste shark

After determining the purpose of the task, researchers sought out existing study/technology related to the goal. The researchers saw the waste shark mega box (WSM) as the best idea to clean up ocean wastes.

Finding ways on how RC boats work

The WSM should be able to move through the ocean waters to collect plastic wastes floating in the ocean. The researchers turn to apply the technology of RC boats to make the WSM swim in the ocean and do the job it was intended to do. The robot should work with a battery voltage of 14.8.

Gathering and buying of materials

The WSM must stay afloat and swim through the ocean. So, the researchers need to find the best material for the body, the automation to make it swim and gather waste, the energy system that will make it work, and the RC to control the WSM.

Measuring components to make sure they fit

The prototype WSM will be smaller for the purposes of this project but should be expandable to a larger prototype as may be required by the actual body of water it will be servicing. The mega box (66.5 cm x 45 cm) (1), fish net (1), metal propeller (1), and servo (1) are the items required for this research. Shafts stainless steel (1), brushless motor (1), ESC (1), and 4s/5s battery (1) were also required for the investigation. The last component is the receiver (1), which is required for the robot to function.



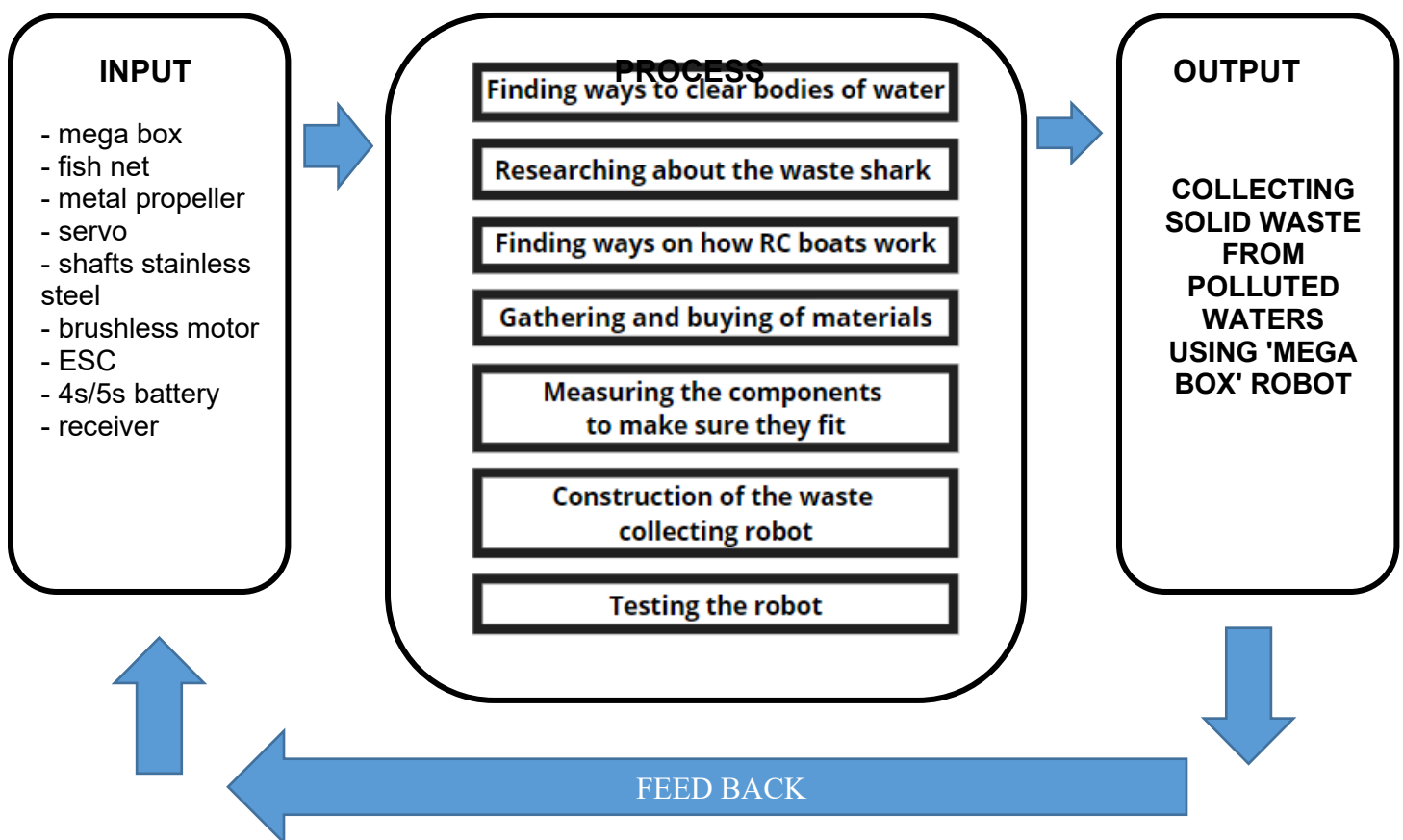
Construction of the waste collecting robot

All materials now available and the measurements done, the team goes to work assembling the robot. The size of the megabox should be 66.5cm x 45cm.

Testing the robot

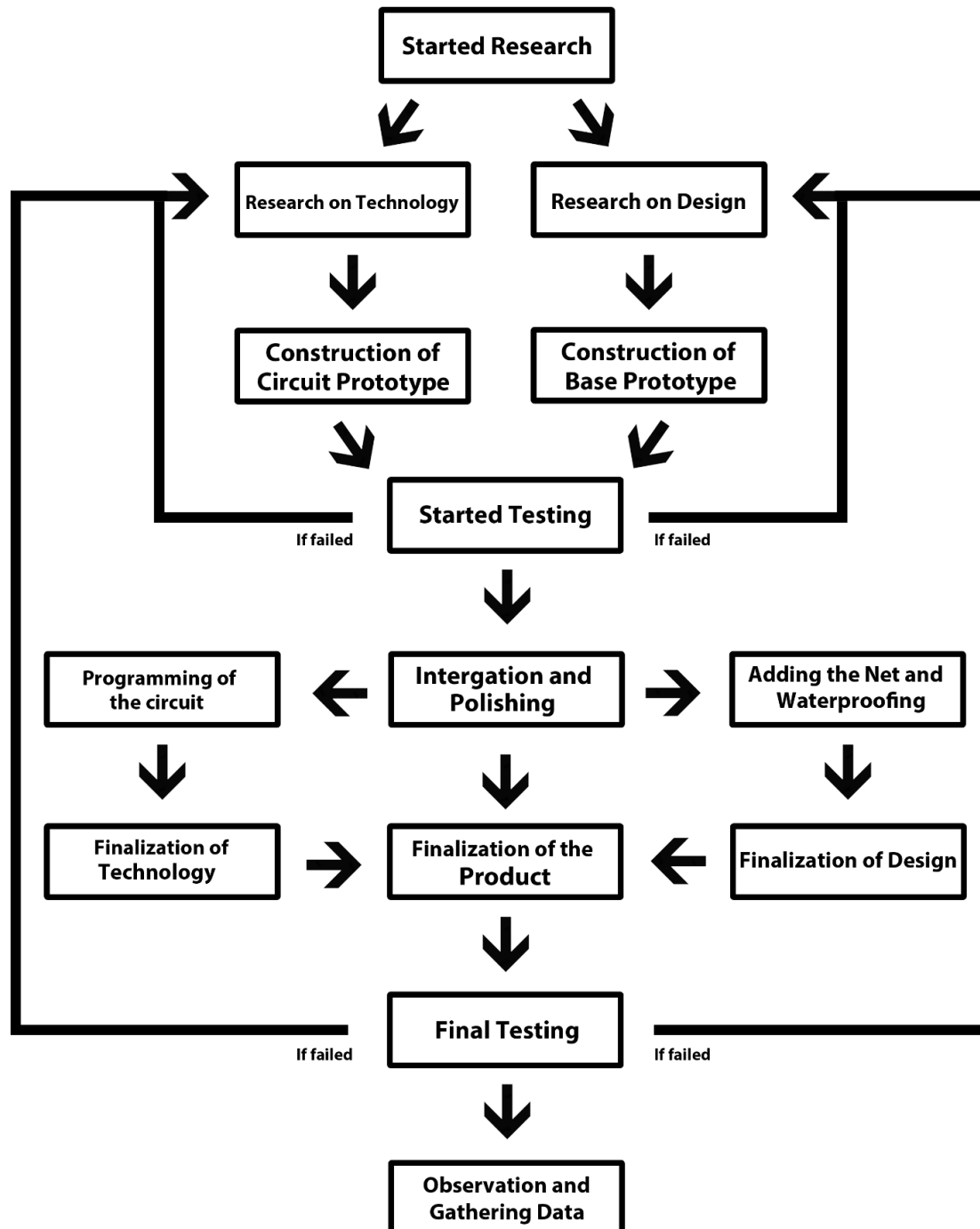
As the project comes to fruition, researchers test the WSM parts work before doing the test runs on water. After checking the functionality of the WSM, researchers will put the project to test in a shallow pool or body of water. The purpose at this point is to see the waste gathering functions at work. As the project works in the shallow pool, the researchers now make the final test of bringing the WSM to an actual river. This final test will not only check its waste gathering capability, but also the RC functions to enable the team to direct the unit to where it should be going to collect wastes and return to deposit the waste collected before it goes through another round. The variables that needs to be tested are the speed (10 kph), weight capacity (50kg), within **maximum constant time of operation which is 2 hours**

Research Framework





SCHEMATIC DIAGRAM



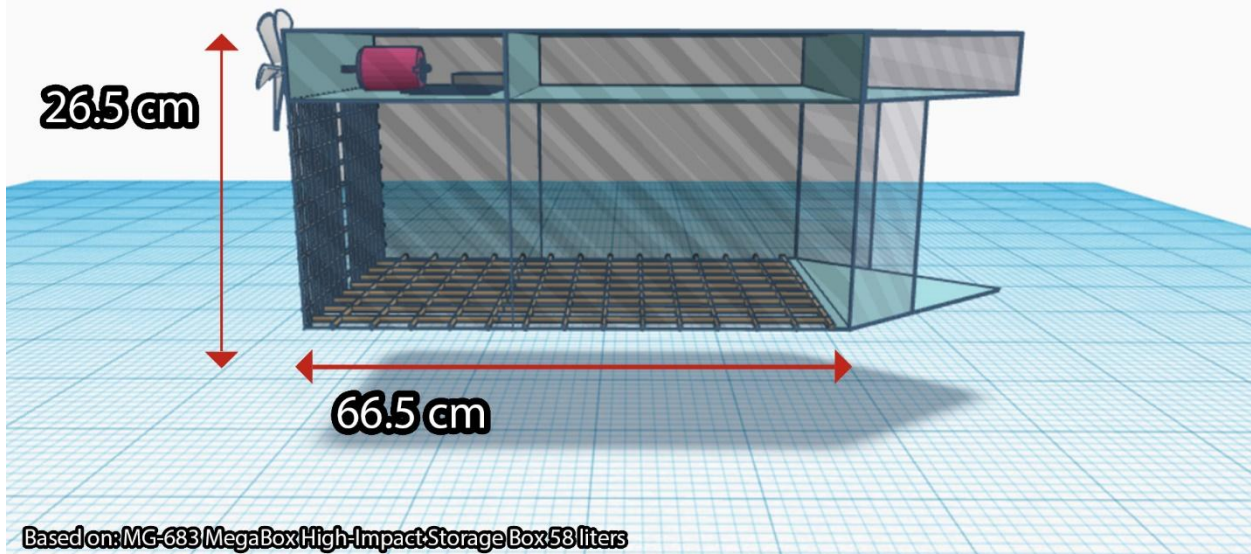
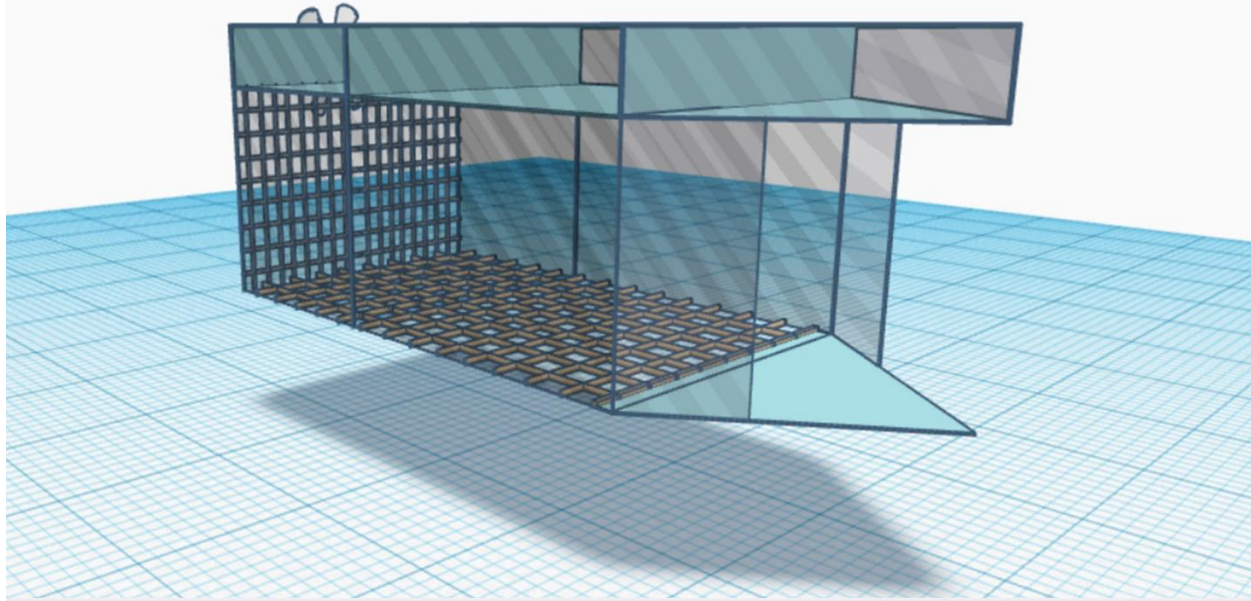


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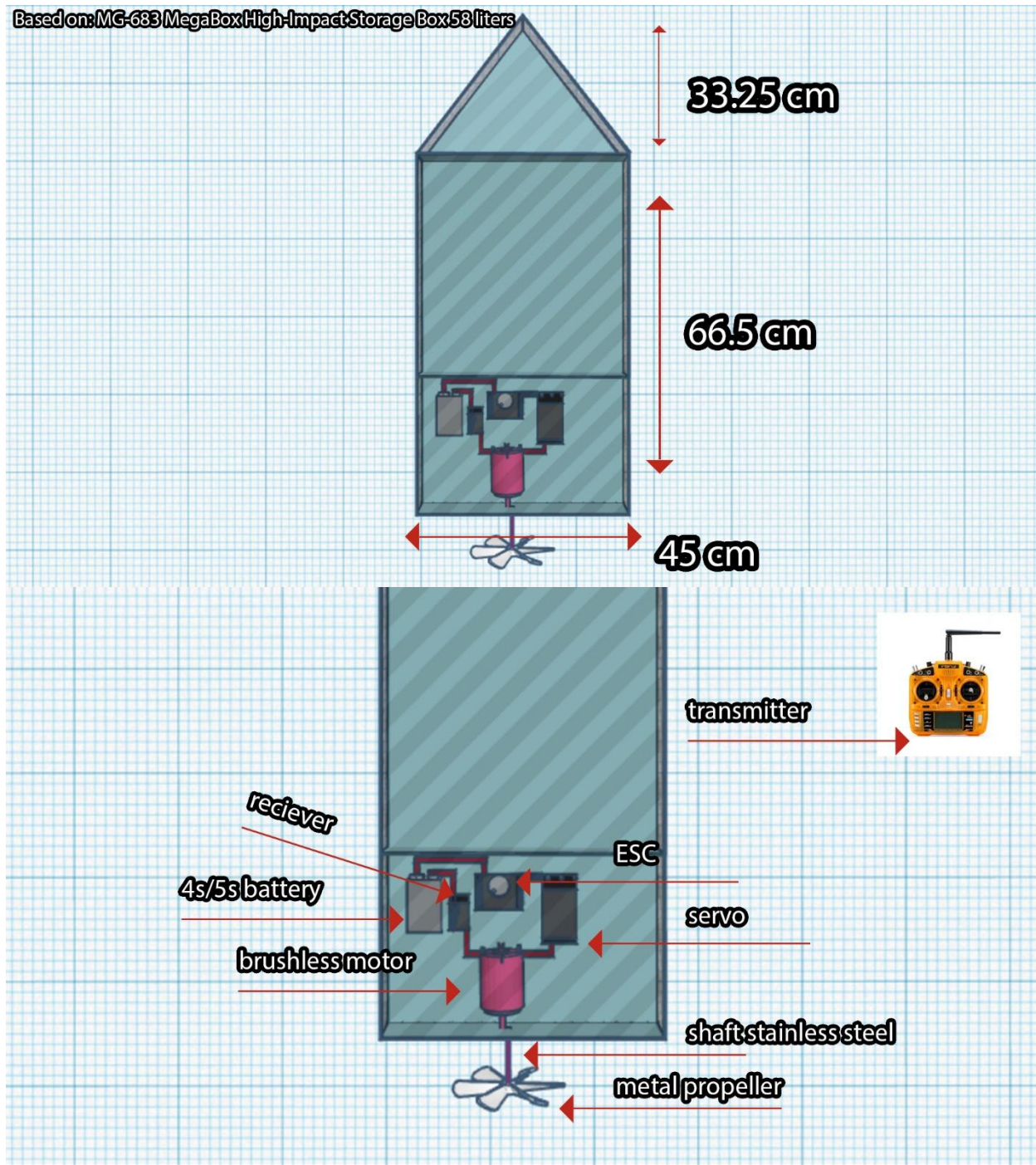


BLUEPRINT





Based on: MG-683 MegaBox High-Impact Storage Box 58 liters





APPENDIX:

Cost Analysis of Investigatory Project

Item	Quantity/ Frequency	Amount	Subtotal (Quantity X Amount)	Location where it will purchased/ undertaken
Mega Box	1	PHP 199.00	PHP 199.00	Lazada
Fishnet	1	PHP 110.00	PHP 110.00	Lazada
Metal Propeller	1	PHP 1,117.00	PHP 1,117.00	Lazada
Servo	1	PHP 200.06	PHP 200.06	Lazada
Shaft Stainless Steel	1	PHP 110.00	PHP 110.00	Lazada
Brushless motor	1	PHP 573.00	PHP 573.00	Lazada
ESC	1	PHP 240.00	PHP 240.00	Lazada
4s/5s Battery	1	PHP 288.00	PHP 288.00	Lazada
Receiver/Transmitter	1	PHP 862.02	PHP 862.02	Lazada
TOTAL PHP 3,699.08 PHP 462.04 Per Student			/ 8 Members	